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35. a) i) Name a reaction for the estimation of Ozone?
ii) Explain lanthanoid contraction and its effects.

(OR)

b) i) Why transition elements exhibit variable oxidation states?
ii) Explain the structure of Ammonia

36. a) i) Give any two characteristics of ionic crystals.
ii) What are the uses of silicones?

(OR)

b) i) What are the differences between order and molecularity?
ii) Write short note on metal excess and metal deficiency defect with an example.

37. a) i) Explain kolbe's reaction
ii) Derive an expression for Ostwald's dilution law

(OR)

b) i) What do you understand by the term Buffer Solution?
ii) How will you distinguish primary, secondary and tertiary alcohols?

38. a) i) Give expression for the hydrolysis constant and degree of hydrolysis of salt of strong acid and strong base
ii) Explain the mechanism of Cannizaro reaction.

(OR)

b) i) Explain
1) Hoff mann's degradation
2) Benzoin condensation
ii) Calculate the percentage efficiency of packing in case of body centered cubic crystal.

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COMMON QUARTERLY EXAMINATION - 2026

Standard XII Reg.No. | | | | | | |

CHEMISTRY

Time : 3.00 hrs **Part - I** ***Marks : 70***

I. Choose the correct answer. **15 x 1 = 15**

1. The gangue removed from Gold during cyanide leaching
a) Ferrous silicate b) Alumino silicate
c) Silicon di oxide d) Calcium silicate
2. Which one of the following reduces tollens reagent
a) formic acid b) acetic acid
c) benzophenone d) none of these
3. _____ is used for the manufacture of petrol by cracking the mineral oils.
a) $AlCl_3$ b) B_2H_6 c) H_3BO_3 d) XeF_2
4. Assertion : Phenol is more reactive than benzene towards electrophilic substitution reaction.
Reason : In the case of phenol, the intermediate arenium ion is more stabilized by resonance.
a) both assertion and reason are true and reason is the correct explanation of assertion.
b) both assertion and reason are true but reason is not the correct explanation of assertion.
c) assertion is true but reason is false
d) both assertion and reason are false.
5. Which is not an hydrogen bonded molecular solid?
a) solid ice b) glucose
c) urea d) methane
6. On reacting with neutral ferric chloride, phenol gives
a) red colour b) violet colour
c) dark green colour d) no colouration
7. A zero order reaction X → Product, with an initial concentration 0.02M has a half life of 10 min. if one starts with concentration 0.06M, then the half life is
a) 30 s
b) 5 min
c) 20 min
d) cannot be predicted using the given information

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8. The pH of an aqueous solution is Zero. The solution is
   a) slightly acidic b) strongly acidic
   c) neutral d) basic
9. An ionic compound AxBy crystallizes in fcc type crystal structure with B ions at the centre of each face and A ion occupying corners of the cube. The correct formula of AxBy is
   a) AB b) AB_3 c) A_3B d) A_8B_6
10. Which one of the following ores is best concentrated by froth – floatation method?
    a) Magnetite b) Haematite
    c) Galena d) Cassiterite
11. Oxidation state of carbon in its hydrides
    a) +4 b) –4 c) +3 d) +2
12. The molarity of given orthophosphoric acid solution is 2M. Its normality is
    a) 6N b) 4N
    c) 2N d) none of these
13. In acid medium, potassium permanganate oxidizes oxalic acid to
    a) oxalate b) carbon dioxide
    c) acetate d) acetic acid
14. The yellow colour in NaCl crystal is due to
    a) excitation of electrons in F centers
    b) reflection of light from Cl–ion on the surface
    c) refraction of light from Na+ ion
    d) all of the above
15. Nitration of Urotropine under controlled condition gives
    a) RDX b) cyclonite
    c) cyclotrimethylene trinitramine d) all the above

Part - II

II. Answer any Six questions. (Q.No.24 is compulsory) **6 x 2 = 12**

16. Define Auto reduction with suitable example.
17. List out the uses of Boron.

18. How will you prepare chlorine by Deacon's process?
19. What are interstitial compounds?
20. Define coordination number. What is the two dimensional coordination number of a molecule in square close packed layer?
21. Write any two tests to identify aldehydes?
22. What are Lewis acids and bases? Give example.
23. How will you prepare acetaldehyde from ethylene glycol?
24. The rate constant for a first order reaction is $1.54 \times 10^{-3}\ s^{-1}$. Calculate its half life time.

Part - III

III. Answer any Six questions. (Q.No.33 is compulsory) **6 x 3 = 18**

25. Describe a method for refining nickel.
26. Discuss the Structure of diborane.
27. What are interhalogen compounds? Give examples.
28. Differentiate Lanthanoids and actinoids?
29. Write the expression for the solubility product of $Ca_3(PO_4)_2$.
30. Write a note on Frenkel defect.
31. Explain pseudo first order reaction with an example.
32. Write the mechanism of Aldol condensation
33. One mole of an organic compound (A) with the formula C_3H_8O, reacts completely with two moles of HI to form compounds (B) and (C). Compound A can be prepared by Williamson synthesis. Identify A, B, C

Part - IV

IV. Answer all the questions. **5 x 5 = 25**

34. a) i) What are the differences between minerals and ores?
    ii) Explain briefly the collision theory of bimolecular reactions

    (OR)

    b) i) Write a short note on Graphene
    ii) Explain zone refining process

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